Qianqian (Bella) He

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SUMMARY

AI/ML engineer and University of Chicago M.S. Computer Science candidate (Dec 2026) with experience building production ML, LLM agents, and distributed data systems spanning 10M+ multimodal and 2M+ time-series data at Alibaba, L'Oréal, and research labs, delivering $8M+ in annual savings. Current focus: software-hardware co-design, agentic AI eval & safety, GPU optimization.

SKILLS

- AI/ML: LLMs, Agentic AI, RAG, RL, Multimodal ML, AI Eval & Safety, Model Training, World Model, NLP, CV, PyTorch
- Engineering: Python, C++, Java, SQL, JavaScript, Go, R, CUDA, Distributed System, API, Docker, Kubernetes, AI Infrastructure
- Data Science: Governance, Modeling, Pipelines, Database, Spark, ETL, Statistics, Experiment Design, Optimization, Forecasting

EDUCATION

The University of Chicago - M.S. in Computer Science, Chicago, IL Sept 2025 - Dec 2026

Relevant Course: Machine Learning, NLP, Computer Vision, Distributed System, Parallel Computing, Advanced C++, Optimization

PROFESSIONAL EXPERIENCE

Argonne National Laboratory – Research Assistant, Chicago, IL Sep 2026 - Present

- Built an energy-aware RTL optimization framework using Academy-based agents to iteratively improve EDA, integrating automated synthesis, functional verification and PPA evaluation to optimize energy efficiency while preserving correctness.
- Developed energy demand prediction and evaluation for BIA project, enabling energy-aware software-hardware co-design.
- Developed AI4HPC agents for scientific software workflows, spanning model inference, training, and optimization.

UChicago Data Science Institute - Research Intern, Chicago, IL June 2026 - Present

- Built QIAO-Bench, a JEPA-based benchmark for state-action decisions and recovery in long-horizon agents, achieving 93.8% final state correctness and 100% rollback correctness.
- Built TraceGuard for provenance-aware agent trajectory analysis, unsafe transition localization, and intervention evaluation, increasing verified attack discovery from 60% to 100% at matched search budget.

Trina Solar Co., Ltd. - Data Scientist, Shanghai, CN Jan 2024 – Aug 2025

- Architected an agentic supply chain optimization system using LLMs, RAG, forecasting, and workflow orchestration across pricing, supplier, logistics, and market intelligence data, achieving $6.65M in annual cost savings.
- Built a production energy management platform integrating CNN-LSTM forecasting, CPLEX optimization, reinforcement learning, real-time data ingestion, and APIs across 2M+ time-series datasets, reducing annual operational costs by $1.53M.
- Built enterprise data governance and analytics foundations, including data catalogs, standards, models and pipelines across supply chain and energy operations to support consistent analysis and decision-making.

L'Oréal S.A. - Data Analytics & Governance Manager, Shanghai, CN Nov 2022 - Dec 2023

- Optimized a transformer-based multimodal recommendation system across 10M+ structured and unstructured datasets, improving recommendation relevance and user satisfaction by 13%.
- Developed a GenAI analytics platform combining ETL automation, natural language querying, predictive analytics, NLP-based insight generation, and dashboards, improving operational efficiency by 21.4%.
- Built multi-channel online/offline data pipelines and a governed analytics platform with standardized data catalogs, domain data models, metadata management, data quality controls, and dashboard analytics.

AISHU Technology Corp. - Product Manager, Shanghai, CN Aug 2021 – Oct 2022

- Led development of an AIOps platform using Java and Rust for anomaly detection, failure prediction, infrastructure monitoring, and root-cause analysis across distributed systems, improving operational efficiency by 30%.
- Designed telemetry processing pipelines and real-time monitoring services supporting servers, networks, and application infrastructure at scale and implemented MLOps and backend orchestration workflows for scalability and reliability.

Alibaba Group Holding Ltd. - Product Manager, Hangzhou, CN May 2020 - July 2021

- Managed development and lifecycle of the Alibaba Tianchi Big Data & AI Platform, improving open dataset access, AI deployment flows, testing, CI/CD, and developer experience with user analytics, rapid prototyping, and workflow orchestration.
- Developed a medical image classification model using ResNet34, FPN and BBN mixup in Python and Pytorch, improving macro F1 by 20.7% for model benchmarking and evaluation.

RESEARCH EXPERIENCE & PUBLICATIONS

Energy Efficiency and Verification of Software-Hardware Co-Design Argonne National Laboratory/Columbia University 2026

- Built energy-aware RTL optimization framework using Academy-based agents to iteratively improve EDA, integrating synthesis, verification, PPA evaluation, and energy efficiency optimization.
- Build agentic system for AI4HPC project, automating the workflows of scientific software development and optimization.
- Build energy prediction and co-design evaluation for BIA project, transforming materials and computer architecture research for energy efficiency.

Evaluation & Safety for Long-Horizon Agents UChicago Data Science Institute/WM@Booth Workshop 2026

- Built QIAO-Bench, a decision point benchmark using JEPA world models for long-horizon agents, enabling state-action prediction and recovery planning, achieving 93.8% recovery-state correctness and 100% rollback correctness.
- Developed TraceGuard, an agent-safety framework using provenance-aware trajectories to pinpoint unsafe transitions, and assess recovery strategies, improving verified attack discovery from 60% to 100% at matched search budget.

SHAP-based Explainable AI for Reliable Energy Consumption Forecasting HKU/EIECT 2024

- Built an end-to-end time series forecasting pipeline using 34,895 train and 1,776 test electricity data, engineered 22 features and compared 5 ML models including XGBoost, LightGBM, random forest, decision tree, and SVM.
- Achieved best performance with XGBoost (MAE 0.0297, MSE 0.00169) and applied SHAP to quantify model-specific feature contributions and improve prediction interpretability and transparency.

Data Augmentation for Breast-Ultrasound Classification NTU Business AI Lab 2023

- Developed a CNN for 3-class breast-ultrasound classification, using 1,578 500x500 images from 600 patients, with 1,103 train/157 valid/318 test images, convolution/max-pooling layers, dropout, and image augmentation.
- Evaluated augmentation strategies, with random rotation and flipping achieving 92.1% test accuracy, versus 91.4% baseline and 89.3% with resize/rescale, demonstrating improved generalization and reduced overfitting.

SELECTED PROJECTS

Agent Security Red-Teaming & Evaluation: Multi-Step Tool Attacks Challenge

- Built an algorithmic red-teaming system for tool-using agents, discovering replayable multi-step attack trajectories across 4 security failure classes and 2 target model families, capturing prompts, memory, tool calls, file/API operations, and policy outcomes for reproducible failure localization and causal analysis.
- Extended safety evaluation toward predictive failure detection and recovery, identifying unsafe-transition boundaries, attributing failures to preceding actions, and determining when agents should continue, verify, replan, or reverse prior actions.

ML Systems & GPU Inference Optimization

- Built an end-to-end neural-network inference and evaluation pipeline with ONNX graph generation and validation, tensor encoding, batched processing, and parallel data loading for reproducible ML workloads. Profiled 4 performance bottleneck classes – memory layout, batching, kernel execution, and CPU/GPU transfers and applied CUDA-aware optimization patterns across inference path.
- Developed experiment configuration, model-validation checks, and reproducible run artifacts to improve evaluation repeatability, systems debugging, and reliability for large-scale model experiments.

Agentic Portfolio Management and Predictive Market Engine

- Architected an event-driven financial platform across 4 service domains—market data, transaction, recommendations, and portfolio management—using ActiveMQ and Apache Camel for asynchronous data and decision workflows.
- Built a market-forecasting and uncertainty layer to generate predictive signals from historical market data, evaluating forecasts with statistical and calibration metrics and incorporating model confidence into portfolio risk and rebalancing decisions.
- Developed LLM-based agentic workflows and tool calling that combine forecasts, market conditions, portfolio state, and investor risk profiles to automate investment analysis, risk evaluation, recommendation, and dynamic rebalancing.

Distributed Energy Simulation & Agent Control Platform

- Developed a distributed energy simulation platform with FastAPI and Docker, exposing authenticated REST APIs and encrypted inter-service communication for state monitoring, scheduling, and energy-control workflows.
- Implemented fault-tolerant coordination through message relaying, automated service re-registration, and hybrid in-memory/relational state management, preserving consistency under distributed service failures.
- Extend the simulator toward an agent RL environment with explicit state, action, transition, and evaluation interfaces, enabling trajectory logging and future work on adaptive control, failure recovery, and reinforcement-learning policies.